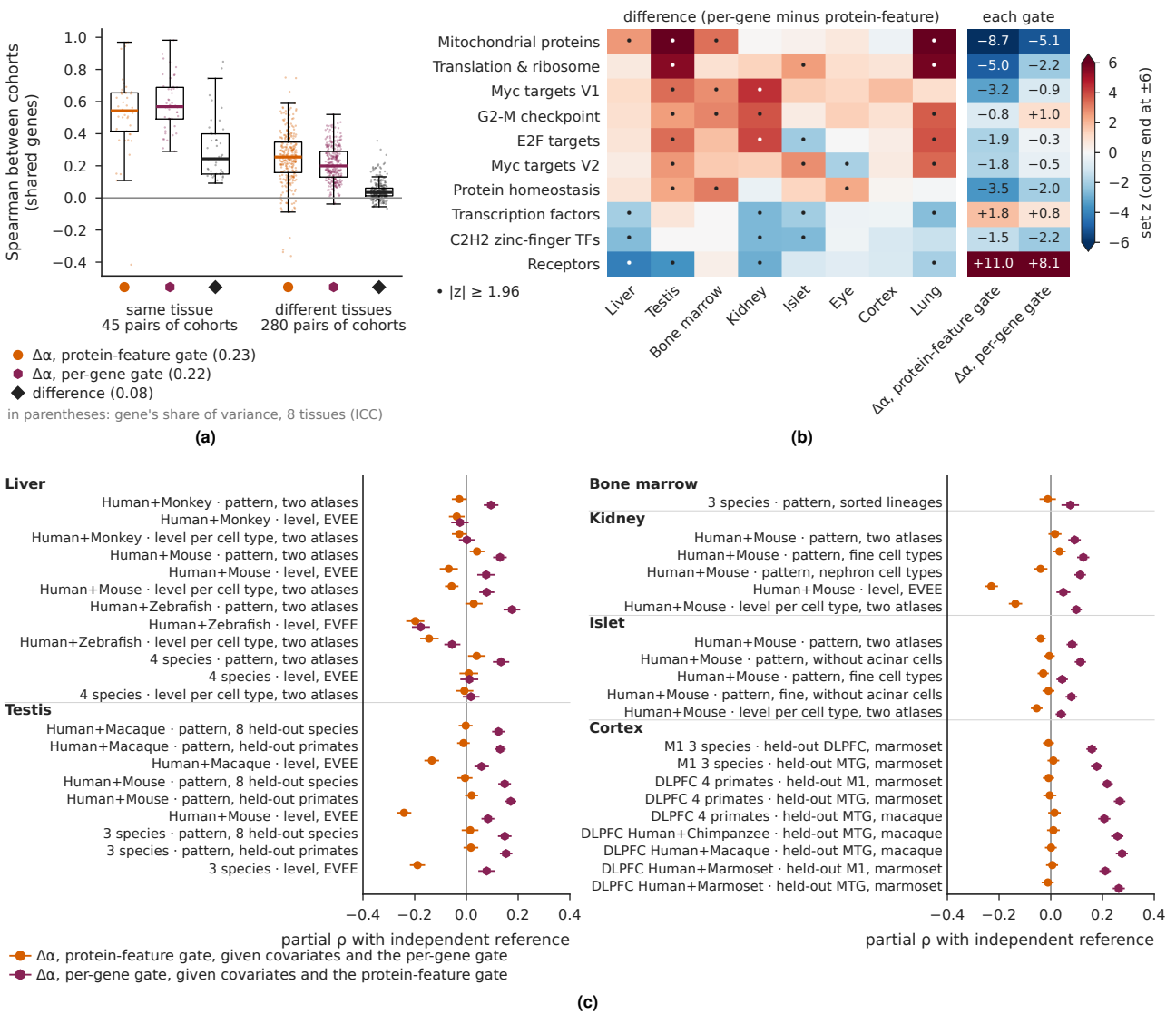

Spearman between cohorts (shared genes)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
same tissue
45 pairs of cohorts
different tissues
280 pairs of cohorts
Δα, protein-feature gate (0.23)
Δα, per-gene gate (0.22)
difference (0.08)
in parentheses: gene's share of variance, 8 tissues (ICC)
(a)
difference (per-gene minus protein-feature)
each gate
Mitochondrial proteins
Translation & ribosome
Myc targets V1
G2-M checkpoint
E2F targets
Myc targets V2
Protein homeostasis
Transcription factors
C2H2 zinc-finger TFs
Receptors
−8.7 −5.1
−5.0 −2.2
−3.2 −0.9
−0.8 +1.0
−1.9 −0.3
−1.8 −0.5
−3.5 −2.0
+1.8 +0.8
−1.5 −2.2
+11.0 +8.1
• |z| ≥ 1.96
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
Δα, protein-feature gate
Δα, per-gene gate
6
4
2
0
−2
−4
−6
set z (colors end at ±6)
(b)
Liver
Human+Monkey · pattern, two atlases
Human+Monkey · level, EVEE
Human+Monkey · level per cell type, two atlases
Human+Mouse · pattern, two atlases
Human+Mouse · level, EVEE
Human+Mouse · level per cell type, two atlases
Human+Zebrafish · pattern, two atlases
Human+Zebrafish · level, EVEE
Human+Zebrafish · level per cell type, two atlases
4 species · pattern, two atlases
4 species · level, EVEE
4 species · level per cell type, two atlases
Testis
Human+Macaque · pattern, 8 held-out species
Human+Macaque · pattern, held-out primates
Human+Macaque · level, EVEE
Human+Mouse · pattern, 8 held-out species
Human+Mouse · pattern, held-out primates
Human+Mouse · level, EVEE
3 species · pattern, 8 held-out species
3 species · pattern, held-out primates
3 species · level, EVEE
−0.4 −0.2 0.0 0.2 0.4
partial ρ with independent reference
Bone marrow
3 species · pattern, sorted lineages
Kidney
Human+Mouse · pattern, two atlases
Human+Mouse · pattern, fine cell types
Human+Mouse · pattern, nephron cell types
Human+Mouse · level, EVEE
Human+Mouse · level per cell type, two atlases
Islet
Human+Mouse · pattern, two atlases
Human+Mouse · pattern, without acinar cells
Human+Mouse · pattern, fine cell types
Human+Mouse · pattern, fine, without acinar cells
Human+Mouse · level per cell type, two atlases
Cortex
M1 3 species · held-out DLPFC, marmoset
M1 3 species · held-out MTG, marmoset
DLPFC 4 primates · held-out M1, marmoset
DLPFC 4 primates · held-out MTG, marmoset
DLPFC 4 primates · held-out MTG, macaque
DLPFC Human+Chimpanzee · held-out MTG, macaque
DLPFC Human+Macaque · held-out MTG, macaque
DLPFC Human+Marmoset · held-out M1, marmoset
DLPFC Human+Marmoset · held-out MTG, marmoset
−0.4 −0.2 0.0 0.2 0.4
partial ρ with independent reference
Δα, protein-feature gate, given covariates and the per-gene gate
Δα, per-gene gate, given covariates and the protein-feature gate
(c)